

CUSTOMER	ACCOUNT / PREMISE	TARIFF CLASSIFICATION	NOTIFIED MAX DEMAND
Impala Plats Rustenburg Mine	7856504676 / 7856504226	Megaflex Diversity (33 kV)	85 740,00 kVA

1. System Architecture & Workflow Pipeline

The Eskom Management Platform operates as a non-lossy 7-stage automated data processing pipeline. It ingests raw utility bills and interval meter readings, reconstructs statutory billing line items, executes multi-variable TOU time-band classification, and flags overcharge recovery claims with full legal citations.

Stage	Module Name	Functional Description	Output Deliverable
1. Ingestion	PDF OCR & Excel Reader	Extracts 17 invoice line items from PDF tax invoices and loads 30-min AMR meter readings (kW, kVA, kVAr, PF).	Non-lossy JSON payload stored in Supabase.
2. Classification	NERSA TOU Engine	Classifies 5,760 interval readings into Peak, Standard, and Off-Peak using published 2025/26 NERSA clock.	Summed TOU energy totals (kWh) per band.
3. Tariff Lookup	Gazette Schedule Matrix	Resolves exact NERSA rates for Megaflex Diversity 33kV ($\geq 500V$ & $< 66kV$ row in Table 3, p.16).	13 distinct published rate constants.
4. Recompute	17-Line Reconciliation Engine	Reconstructs every billed line item from first principles ($\$Quantity \times Rate\$$) to check for variances.	Itemized variance table (0% match to cent).
5. Audit Engine	Overcharge Recovery Module	Applies 6 mathematical audit rules (TX rate, curtailment ratchet, pro-rata split, wheeling subsidy netting).	Itemized Recovery Register & Claim Cards.
6. Export	Dispute Package Exporter	Generates CSV dispute packages protected against formula injection, executive PDFs, and raw JSON logs.	Formatted Eskom dispute filing package.
7. Database	Supabase PostgreSQL RLS	Persists multi-tenant accounts, invoices, telemetry, and recovery records under Row-Level Security.	Enterprise cloud database reflection.

2. Data Provenance & Location of Numbers

All calculations on the platform originate from three authoritative data sources:

- Source 1: Eskom Utility Tax Invoices** — Scanned PDF bills (e.g., Page 5 of Invoice #785101497007, #7856504676, #785684906677, #785595072130) providing Account #, Premise ID, NMD, and 17 invoiced line items.
- Source 2: AMR Meter Telemetry** — TrendExport_Millennium 33kV workbook containing 5,760 30-minute interval readings of kW active power, kVA apparent power, kVAr reactive power, and Power Factor.
- Source 3: Statutory Tariff Gazette** — NERSA-approved Eskom Schedule of Standard Prices 2025/26 (Table 3, p.16 for $\geq 500V$ & $< 66kV$) and 2026/27 (Table 1, p.14).

3. Notified Maximum Demand (NMD) & Capacity Formulas

Notified Maximum Demand (NMD) is the contracted capacity ceiling reserved for the supply point (**85 740,00 kVA**). Under NERSA Megaflex Schedule Page 2, **Annual Utilised Capacity** is defined as:

Annual Utilised Capacity (kVA) = max(Notified Maximum Demand, Highest Peak Demand Recorded in Rolling 12 Months)

Four capacity and demand charges ride on kVA volumes under Megaflex Diversity 33kV (Table 3, p.16):

1. Distribution Network Capacity = Utilised Capacity (85 740 kVA) × R 35.98 / kVA / month = R 3 084 925.20
2. Transmission Network Capacity = Utilised Capacity (85 740 kVA) × R 10.25 / kVA / month = R 878 835.00
3. Generation Capacity = Utilised Capacity (85 740 kVA) × R 8.09 / kVA / month = R 693 636.60
4. Distribution Network Demand = Billed Peak (86 432.57 kVA) × R 24.17 / kVA / month = R 2 089 075.22

Demand Penalty Rationale: Combined capacity ceiling rate is **R 54.32 / kVA / month** (R 35.98 + R 10.25 + R 8.09). A 1,000 kVA spike above NMD incurs **R 54 320 / month**, or **R 651 840 before it rolls off over 12 months**.

4. The 17-Line Invoice Reconciliation Engine

Reconstruction of the February 2026 Eskom Tax Invoice (#785101497007) from first principles:

Billed Line Item	Metered / Contract Qty	NERSA Rate (Table 3, p.16)	Calculated Amount (ex VAT)	Invoiced Amount (ex VAT)	Status
Low Season Peak Energy	6 401 924.40 kWh	276.78 c/kWh (R 2.7678)	R 17 719 245.25	R 17 719 245.25	MATCH (0%)
Low Season Standard Energy	19 432 557.60 kWh	155.62 c/kWh (R 1.5562)	R 30 240 946.76	R 30 240 946.76	MATCH (0%)
Low Season Off-Peak Energy	23 429 967.60 kWh	111.15 c/kWh (R 1.1115)	R 26 042 409.43	R 26 042 409.43	MATCH (0%)
Legacy Charge	49 264 449.60 kWh	22.20 c/kWh (R 0.2220)	R 10 936 707.81	R 10 936 707.81	MATCH (0%)
Network Capacity Charge	85 740.00 kVA	R 35.98 / kVA	R 3 084 925.20	R 3 084 925.20	MATCH (0%)
Electrification Subsidy	49 264 450.00 kWh	4.94 c/kWh (R 0.0494)	R 2 433 663.83	R 2 433 663.83	MATCH (0%)
Affordability Subsidy	49 264 450.00 kWh	4.69 c/kWh (R 0.0469)	R 2 310 502.71	R 2 310 502.71	MATCH (0%)
Network Demand Charge	86 432.57 kVA	R 24.17 / kVA	R 2 089 075.22	R 2 089 075.22	MATCH (0%)
Transmission Network Capacity	85 740.00 kVA	R 10.25 / kVA	R 878 835.00	R 878 835.00	MATCH (0%)
Generation Capacity Charge	85 740.00 kVA	R 8.09 / kVA	R 693 636.60	R 693 636.60	MATCH (0%)
Ancillary Service Charge	49 264 450.00 kWh	0.39 c/kWh (R 0.0039)	R 192 131.36	R 192 131.36	MATCH (0%)
Service Charge (Key Customer)	31 days	R 1 118.46 / day	R 34 672.26	R 34 672.26	MATCH (0%)
Administration Charge	31 days	R 19.37 / day	R 600.47	R 600.47	MATCH (0%)
Subtotal (13 Rate Lines)	—	—	R 96 657 351.90	R 96 657 351.90	100% MATCH
Connection Charges (4 Lines)	Contractual	—	Unsourced	R 351 887.21	CONTRACTUAL FEE
Total Invoiced (ex VAT)	—	—	—	R 97 009 239.11	EXACT BILL MATCH

5. Published 2025/26 Time-Of-Use (TOU) Engine

The platform applies the published 2025/26 Eskom NERSA TOU Clock (Schedule §3.2, p.8):

- Low Season Weekday Peak: 06:00–08:00 & 17:00–20:00
- Low Season Weekday Standard: 08:00–17:00 & 20:00–22:00
- Low Season Weekday Off-Peak: 22:00–06:00
- Saturday Standard: 07:00–12:00 & 18:00–20:00
- Sunday Standard: 18:00–20:00 (New 2025/26 Sunday Window)

Public Holiday Statutory Rule (§10, p.11): Only 5 public holidays (New Year's Day, Good Friday, Family Day, Christmas Day, Day of Goodwill) are billed as Sunday. All other public holidays (Human Rights Day, Freedom Day, Workers' Day, Youth Day, Women's Day, Heritage Day, Day of Reconciliation) are billed on the Saturday profile.

6. Overcharge Recoveries Audit Engine & Formulas

Period	Dispute Category	Invoiced (ex VAT)	Recovery Claim	Audit Status & Legal Citation
Feb 2026	Transmission Capacity Rate Check	R 97 009 239.11	R 878 835.00	BLOCKED / CONTRACTUAL ASK Table 3 prices TX at R10.25/kVA for 33kV; requires Connection Contract to claim zero rate.
Mar 2026	Curtailment Demand Ratchet Reversal	R 98 380 358.13	R 601 365.00	BLOCKED / CONTROL ROOM TIMESTAMPS Spike during mandatory curtailment window; requires System Operator log.
Apr 2026	Pro-Rata Tariff-Year Split	R 91 251 855.72	R 620 450.40	BUILT & VERIFIED TO CENT 13d/16d tariff-year split matches to the exact cent.
May 2026	Renewable Wheeling Subsidy Netting	R 97 169 250.00	R 318 000.00	READY / PPA CLAUSE CHECK Wheeling solar netting check on gross vs net electrification subsidy.

Pro-Rata Tariff-Year Split Formula (April 2026):

Pro-Rata Rate = (13 / 29) × Rate_2025/26 + (16 / 29) × Rate_2026/27

Wheeling Subsidy Netting Formula:

Recovery = (Gross Intake kWh – Net Import kWh) × R 0.0537 / kWh